8.6.2 Ergodicity

Following the discussion in [220], let us show ergodicity for the stochastic isokinetic harmonic oscillator:

$$\begin{aligned} \mathrm{d}q &= p\mathrm{d}t \\ \mathrm{d}p &= [-q - \lambda p]\mathrm{d}t \\ \mathrm{d}\xi_1 &= [-\lambda\xi_1 - \xi_2\xi_1]\mathrm{d}t \\ \mathrm{d}\xi_2 &= [\xi_1^2 - \mathrm{k_B T} - \gamma\xi_2]\mathrm{d}t + \sigma\mathrm{dW} \end{aligned} \tag{8.45}$$

with $\lambda = \frac{-qp-\xi_2\xi_1^2/2}{p^2+\xi_1^2/2}$ and $\sigma = \sqrt{2\gamma \mathrm{k_B T}}$. We have taken the coupling coefficients μ_1, μ_2 both one for simplicity as their particular values (as long as nonzero) have no bearing on the ergodic property.

The constraint, $p^2 + \xi_1^2/2 = \text{const}$, defines a three-dimensional manifold $\mathcal{M}$, a generalized cylinder. $\xi_1 = 0$ is an invariant manifold of the isokinetic equations, with solutions therefore confined for all time to one or the other half-space ($\xi_1 < 0$ or $\xi_1 > 0$). It is enough to sample one or the other of the two half spaces. To show that the solutions do so, we need to demonstrate that an appropriate Hörmander condition holds. Define $\boldsymbol{f}$ and $\boldsymbol{g}$ as the vector fields

$$\boldsymbol{f} = p\partial_q - (q + \lambda p)\partial_p - (\lambda\xi_1 + \xi_2\xi_1)\partial_{\xi_1} + \left[(\xi_1^2 - \mathrm{k_B T}) - \gamma\xi_2\right]\partial_{\xi_2}, \tag{8.46}$$

and

$$\boldsymbol{g} = \sigma\partial_{\xi_2}. \tag{8.47}$$

We will show that the dimension of the ideal spanned by the iterated commutators of the vector fields $\boldsymbol{f}$ and $\boldsymbol{g}$ is three, i.e. that it spans the tangent space to $\mathcal{M}$ at any point. For this purpose, we use the collection of vectors

$$\boldsymbol{f}, \boldsymbol{g}, [\boldsymbol{f}, \boldsymbol{g}], [\boldsymbol{f}, [\boldsymbol{f}, \boldsymbol{g}]], [\boldsymbol{f}, [\boldsymbol{f}, [\boldsymbol{f}, \boldsymbol{g}]]].$$

We will see that these provide a basis at every point excluding the one-dimensional set defined by the union $\mathcal{M}_0$ of the two lines ℓ_+, ℓ_- defined by

$$q = 0, \qquad p = 0, \qquad \xi_1 = \pm\sqrt{2\mu_1^{-1}\mathrm{k_B T}},$$

and arbitrary ξ_2. [Note: the choice $\xi_1 = \pm\sqrt{2\mu_1^{-1}\mathrm{k_B T}}$ is made so that the isokinetic constraint is satisfied. In [220] this value was incorrectly defined.] The vectors $\boldsymbol{f}$ and $\boldsymbol{g}$ are linearly independent except on $\mathcal{M}_0$.

The proof requires calculation of the vector fields

$$\boldsymbol{u}_0 = \boldsymbol{\Pi f}, \qquad \boldsymbol{u}_1 = \boldsymbol{\Pi}\,[\boldsymbol{f},\boldsymbol{g}]\,, \qquad \boldsymbol{u}_2 = \boldsymbol{\Pi}\,[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]], \qquad \boldsymbol{u}_3 = \boldsymbol{\Pi}\,[\boldsymbol{f},[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]]],$$

where $\boldsymbol{\Pi}$ is the projector onto the first three components. Because $\boldsymbol{g} = \sigma \boldsymbol{e}_4$, it is enough to show that any pair from among these four vectors are linearly independent, except possibly on $\mathcal{M}_0$. Note that projection can only be performed after the full calculation of the relevant commutator. Because the commutators quickly become quite messy, we used a symbolic computing package to assist in these calculations.

The first two vectors are:

$$\mathbf{u}_0 = p\partial_q - (q + \lambda p)\partial_p - (\lambda\xi_1 + \xi_1\xi_2)\partial_{\xi_1}, \tag{8.48}$$

and

$$\mathbf{u}_1 = \sigma p \frac{\partial\lambda}{\partial\xi_2}\partial_p + \xi_1\left(1 + \frac{\partial\lambda}{\partial\xi_2}\right)\partial_{\xi_1}. \tag{8.49}$$

After expanding the derivatives of λ we have

$$\mathbf{u}_1 = p\partial_q + \xi_1^2\left(\frac{-q + p\xi_2}{D}\right)\partial_p - 2\xi_1 p\left(\frac{-q + p\xi_2}{D}\right)\partial_{\xi_1}, \tag{8.50}$$

and

$$\mathbf{u}_2 = -\sigma(\xi_1^2 p/D)\partial_p + (2\sigma\xi_1 p^2/D)\partial_{\xi_1}, \tag{8.51}$$

where $D = 2p^2 + \xi_1^2$.

If $q \neq 0$ and $p \neq 0$, then clearly $\mathbf{u}_1$ and $\mathbf{u}_2$ are linearly independent. However, if $p = 0$ then $\mathbf{u}_2 = 0$. We proceed to compute the commutator $[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]]$. Projecting to the first three components results in

$$\mathbf{u}_2 = \sigma\left(\frac{\xi_1^2 p}{D}\right)\partial_q - \sigma\left(\frac{2\gamma p^3 + \gamma\xi_1^2 p - \xi_1^2 q}{D^2}\right)\partial_p + 2\sigma\xi_1 p\left(\frac{2\gamma p^3 + \gamma\xi_1^2 p - \xi_1^2 q}{D^2}\right)\partial_{\xi_1}. \tag{8.52}$$

Again, we substitute $p = 0$ and find that

$$\mathbf{u}_2|_{p=0} = (\sigma\xi_1^2 q/D^2)\partial_p. \tag{8.53}$$

Unfortunately, this is parallel to $\mathbf{u}_1$ (for $p = 0$), and hence yet one more commutator is required in this case.

We calculate $[\boldsymbol{f},[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]]]$ and project onto the space spanned by the first three components to find

$$\mathbf{u}_3 = \sigma\left(\frac{2\gamma p^3 - 4p^3\xi_2 + 4p^2 q + \xi_1^2 p\xi_2 + \gamma\xi_1^2 p - 2\xi_1^2 q}{D^2}\right)\partial_q$$

$$-\sigma\xi_1^2\left(\frac{\eta}{D^3}\right)\partial_p + 2\sigma\xi_1 p\left(\frac{\eta}{D^3}\right)\partial_{\xi_1}. \tag{8.54}$$

with

$$\begin{aligned}\eta = {} & 4p^5\gamma^2 - 8p^5\xi_1^2 - 4\xi_1^2 p^3 + 4p^3\gamma^2\xi_1^2 - 4p^3\xi_1^4 \\ & - 2\gamma\xi_1^2 p^2 q + 4p^2 q\xi_2\xi_1^2 - 4\xi_1^2 q^2 p - 2\xi_1^4 p + p\gamma^2\xi_1^4 - q\gamma\xi_1^4 + \xi_1^4\xi_2 q.\end{aligned} \tag{8.55}$$

Along $p = 0$, $\mathbf{u}_3$ becomes

$$\mathbf{u}_3|_{p=0} = -2\sigma q\partial_q - \sigma\left(\frac{-q\gamma\xi_1^4 + \xi_1^4\xi_2 q}{\xi_1^4}\right)\partial_p$$

$$= -2\sigma q\partial_q - \sigma q(-\gamma + \xi_2)\partial_p. \tag{8.56}$$

Together with $\mathbf{u}_0$ we obtain a linearly independent set as long as $q \neq 0$.

If $p = 0$ and $q = 0$ then we are on $\mathcal{M}_0$. If $p \neq 0$ then $\boldsymbol{u}_2$ and $\boldsymbol{u}_0$ are linearly independent. If $q \neq 0$ then $\boldsymbol{u}_3$ and $\boldsymbol{u}_0$ are linearly independent. From these we establish that the Hörmander condition holds everywhere except on a space of co-dimension 2 (that is except on the one-dimensional set $\mathcal{M}_0$).

In this case, it is possible to complete the proof by introducing a Lyapunov function ϕ to demonstrate the necessary contractivity in the ξ_2 direction. The generator of this system is

$$\mathcal{L}_{\text{isok}} = p\frac{\partial}{\partial q} + \left[F - (p^2/\Lambda)F\right]\frac{\partial}{\partial p} - \frac{p\xi_1 F}{\Lambda}\frac{\partial}{\partial\xi_1} + \mathcal{L}_{\text{N}} + \mathcal{L}_{\text{O}}$$

with $\Lambda = p^2 + \xi_1^2/2$,

$$\mathcal{L}_{\text{N}} = \frac{1}{2}\frac{\xi_1^2\xi_2 p}{\Lambda}\frac{\partial}{\partial p} + \frac{1}{2}\frac{\xi_1^3\xi_2}{\Lambda}\frac{\partial}{\partial\xi_1} - \xi_2\xi_1\frac{\partial}{\partial\xi_1} + [\xi_1^2 - \beta^{-1}]\frac{\partial}{\partial\xi_2},$$

and

$$\mathcal{L}_{\text{O}} = -\gamma\xi_2\frac{\partial}{\partial\xi_2} + \frac{\sigma^2}{2}\frac{\partial^2}{\partial\xi_2^2}.$$